

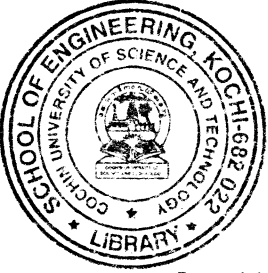
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**B.Tech. Degree VIII Semester Supplementary Examination
July 2019**

**ME 15-1801 COMPRESSIBLE FLUID FLOW
(2015 Scheme)**

Time : 3 Hours

Maximum Marks : 60



**PART A
(Answer ALL questions)**

(10 × 2 = 20)

- I. (a) Derive an expression for acoustic velocity as a function of temperature, ratio of specific heats and characteristic gas constant.
 (b) Differentiate between stagnation state and critical state.
 (c) Derive an expression for the maximum mass flow rate through a nozzle.
 (d) The pressures on either side of a normal shock wave are 100 kPa and 400 kPa. Find by using data book the
 (e) Explain how the analysis for a normal shock wave can be used for the analysis of an oblique shock wave.
 (f) Prove that the Mach number at the maximum entropy point on a Fanno curve is unity.
 (g) Differentiate between thermal choking and friction choking.
 (h) Write a short note on Prandtl Pitot static probe.
 (i) Derive an expression for Temperature Recovery Factor.
 (j) What is a hot wire anemometer?

PART B

(4 × 10 = 40)

- II. (a) Derive an expression for the variation of pressure in the troposphere. (4)
 (b) Describe the behaviour of flow through a convergent-divergent nozzle for various back pressures. (6)

OR

- III. (a) Show that for sonic flow the deviation between the compressible and incompressible flow values of the pressure coefficient of a perfect gas with $\gamma=1.4$ is about 27.5%. (5)
 (b) The pressure, temperature and Mach number at the entry of a flow passage are 2.45 bar, 26.5°C and 1.4 respectively. If the exit Mach number is 2.5, determine for isentropic flow of perfect gas ($\gamma=1.3$, $R=0.469$ kJ/kg-K): (5)
 (i) Stagnation temperature
 (ii) Temperature and velocity of gas at exit
 (iii) The flow rate per square meter of inlet cross-section



(P.T.O.)

- IV. An aircraft engine employs a subsonic inlet diffuser of area ratio 4. Free stream air at a total pressure and temperature of 1 bar and 570 K approaches the diffuser with a Mach number 2.2. A shock wave stands just outside the diffuser inlet. Determine Mach number, temperature and pressure at exit of diffuser and loss in stagnation pressure. (10)

OR

- V. (a) Air having a Mach number 3, approaches a symmetrical wedge having a wedge angle of 30° . The pressure and temperature of air are 1 bar and 27°C . Find the Mach number and velocity of flow downstream of the shock wave, assuming that a weak oblique shock is formed. Also find the downstream pressure and temperature. (6)
- (b) Air at a pressure of 50 kPa and a temperature of 300 K is to be isentropically expanded from a Mach number of 1.5 to achieve a pressure of 20 kPa. Determine the flow deflection angle required. Also determine the final Mach number and temperature of gas. (4)
- VI. Air at a stagnation pressure of 10 bar and stagnation temperature of 400 K is supplied to a duct of 50 mm diameter. The Fanning coefficient of friction ' f ' for the duct surface is 0.002. If the Mach number changes from 3.0 at the entry to 1.0 at the exit, determine (i) the length of the duct (ii) the mass flow rate. (10)

OR

- VII. The pressure, temperature and velocity of a gas in a combustor at entry are 0.343 bar, 310 K and 60 m/s. Determine Mach number, pressure, temperature and velocity at exit of the combustor, if the increase in stagnation enthalpy of the gas due to heat addition is 1172.5 kJ/kg. Assume $\gamma = 1.4$ and $C_p = 1005 \text{ J/kgK}$. (10)
- VIII. (a) Explain the working of a shock tube with a neat figure. (5)
- (b) With a neat figure explain the arrangement adopted in Schlieren technique. (5)

OR

- IX. (a) With the help of a neat sketch explain the working of a Mach-Zehnder interferometer used in the measurement of fluid density. (5)
- (b) Explain the working of an open circuit supersonic wind tunnel with the help of a neat figure. (5)
